Microservices Resiliency

What is Resilience?

In distributed systems, **anything can happen**: network failures, service crashes, slow responses, or unexpected load spikes. (*Remember the Fallacies of Distributed Systems!*)

Resilience is the system's ability to:

- **Recover from failures**
- **Continue operating** despite partial failures
- **Prevent cascading failures** across services

> *An issue in one system can propagate and **affect the behavior and performance of others**.*

> *Think of resilience as a combination of **fault-tolerance**, **graceful degradation**, and **self-healing**.*

State of Resilience 2025 Survey

- **95%** aware of operational weaknesses
- **84%** lost ≥$10,000 due to outages
- **55%** experienced weekly disruptions
- **39%** reported employee burnout from outages

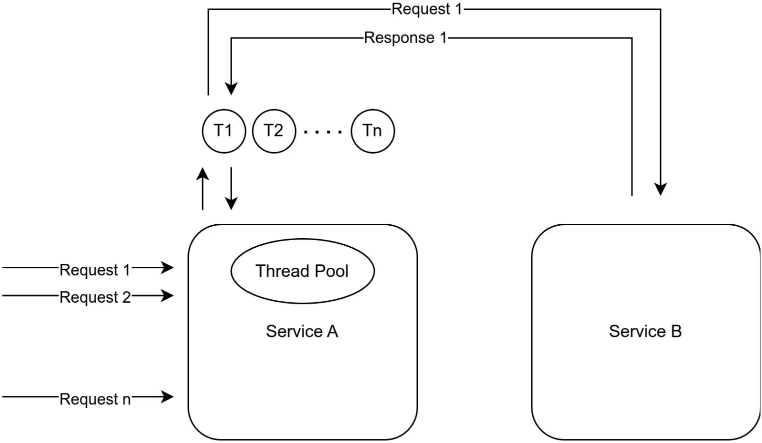
> *Outages are frequent, costly, and damaging to both business and staff morale.*

Thread-Related Risks

One Thread Per Request

Traditional synchronous model:

- Service A assigns a **dedicated thread** to handle each request to Service B.
- The thread is **blocked while waiting**, consuming memory and some CPU. (Spring Boot Starter Web) enables this model.


Request 1
Response 1
T1
T2
Tn
Request 1
Request 2
Request n
Thread Pool
Service A
Service B

Problems at scale:

1. **Memory consumption:** Each thread uses ~2MB; thousands of concurrent requests quickly consume gigabytes.

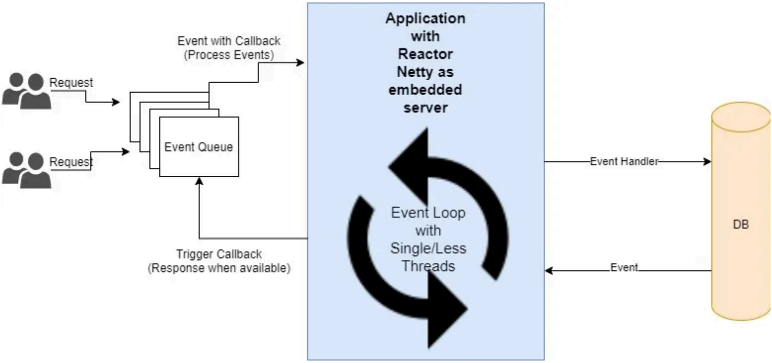
2. **Thread pool saturation:** Once the pool (`server.tomcat.threads.max`) is full, new requests queue → increased latency → potential rejection (DoS).
3. **Resource inefficiency:** Idle threads still consume CPU to manage scheduling, leaving less CPU for actual work.

> *This model works for low-to-medium traffic but can **collapse under high load or slow downstream services**.*

Reactive Programming

Idea: Handle requests asynchronously with **non-blocking I/O**.

- Operations like DB queries or HTTP calls **don't block threads**.
- Instead, a **callback** resumes the operation once data is ready. (Spring Boot WebFlux, Quarkus, FastAPI)


Application with Reactor Netty as embedded server
Event with Callback (Process Events)
Request
Request
Event Queue
Event Loop with Single/Less Threads
Event Handler
Event
DB
Trigger Callback (Response when available)

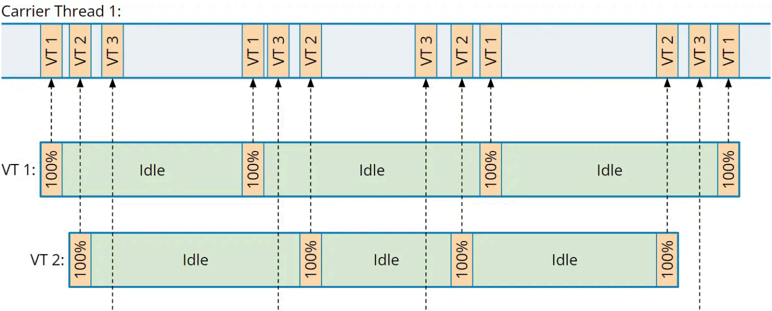
Key benefits:

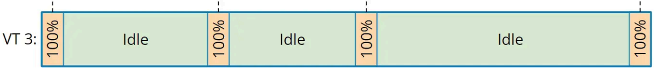
- **High concurrency** → can handle thousands of requests on a single thread
- **Better CPU utilization** → threads do actual computation instead of waiting (co-routines as used for waiting)

Java Virtual Threads

A modern alternative to both thread-per-request and reactive models.

- Lightweight threads managed by the **JVM**, not the OS.
- Traditional **blocking code can be used**, but at massive scale.


Carrier Thread 1:
VT 1
VT 2
VT 3
VT 1
VT 3
VT 2
VT 3
VT 2
VT 1
VT 2
VT 3
VT 1
VT 1:
100%
Idle
100%
Idle
100%
Idle
100%
VT 2:
100%
Idle
100%
Idle
100%
Idle
100%


VT 3:
100%
Idle
100%
Idle
100%
Idle
100%

Key features:

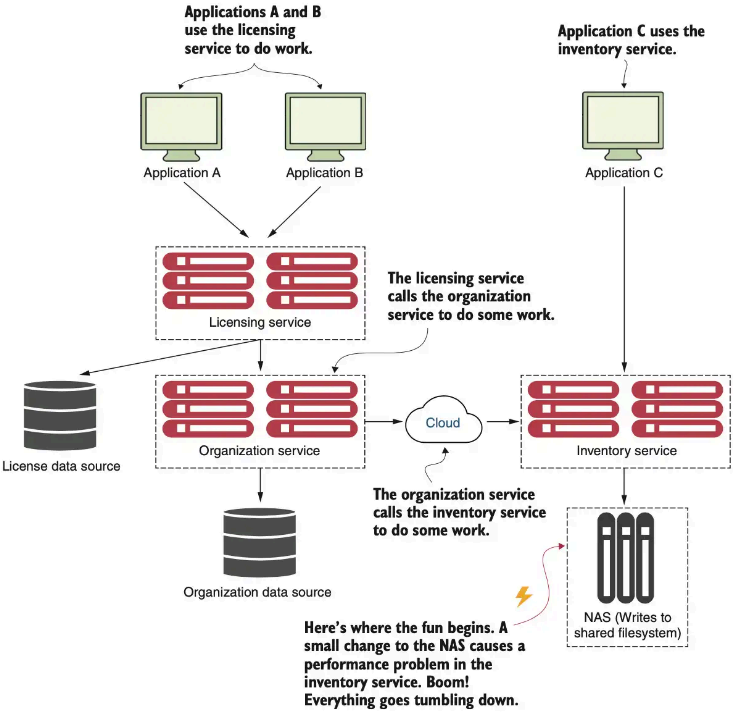
- **Cheap to create and manage:** thousands of virtual threads possible with minimal overhead.
- **Simplifies code:** no callback chains or reactive frameworks needed.
- **Compatible with existing libraries:** works with blocking I/O libraries without rewriting code.

> *Best for teams who want **simplicity** + **high concurrency**, without learning reactive paradigms.*

Real-World example

A minor NAS configuration change caused:

1. Inventory service reads slowed dramatically
2. Organization service threads and DB connections were **exhausted**
3. Licensing service calls queued → **cascading failure across three services**


Applications A and B use the licensing service to do work.

Application C uses the inventory service.

The licensing service calls the organization service to do some work.

The organization service calls the inventory service to do some work.

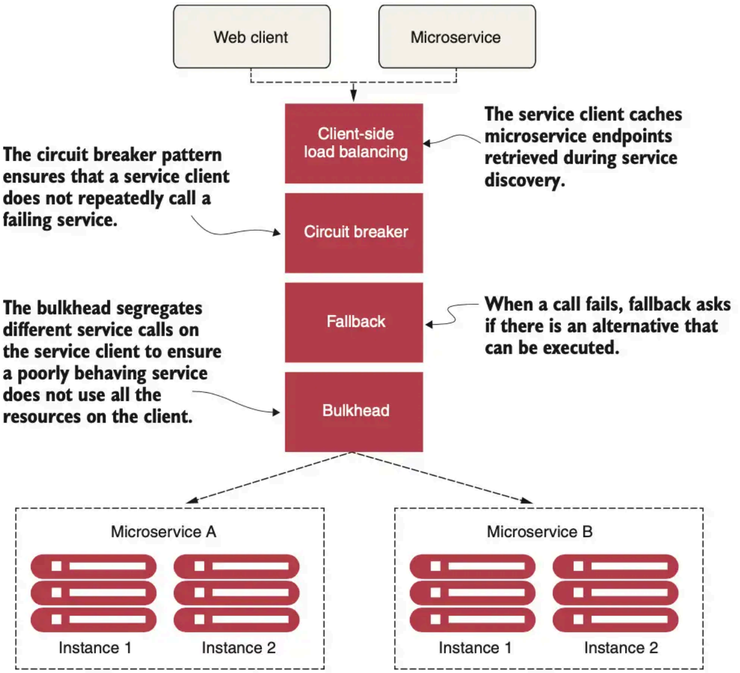
Here's where the fun begins. A small change to the NAS causes a performance problem in the inventory service. Boom! Everything goes tumbling down.

Application A
Application B
Application C
Licensing service
License data source
Organization service
Cloud
Inventory service
Organization data source
NAS (Writes to shared filesystem)

> *Single slow service → ripple effect → entire application ecosystem impacted.*

Client-Side Resiliency Patterns

Protect **clients** from remote service failures.


Web client
Microservice
Client-side load balancing
Circuit breaker
Fallback
Bulkhead
Microservice A
Instance 1
Instance 2
Microservice B
Instance 1
Instance 2

The circuit breaker pattern ensures that a service client does not repeatedly call a failing service.

The service client caches microservice endpoints retrieved during service discovery.

When a call fails, fallback asks if there is an alternative that can be executed.

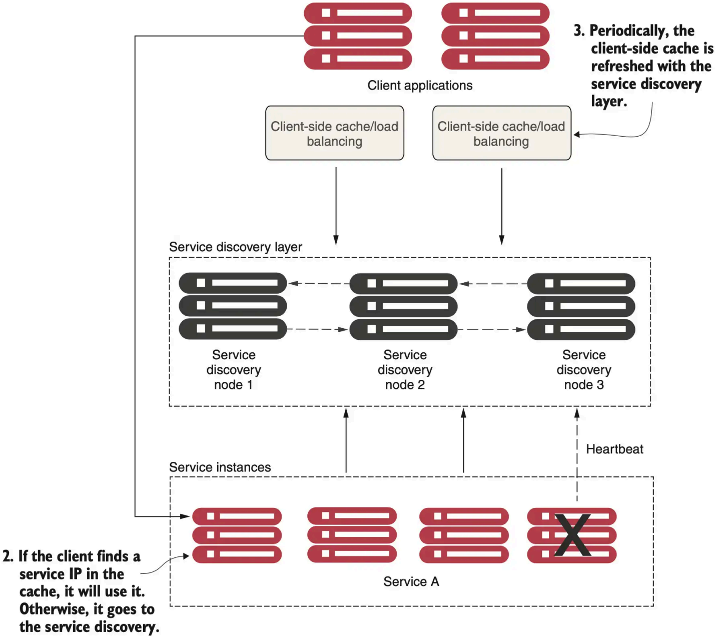
The bulkhead segregates different service calls on the service client to ensure a poorly behaving service does not use all the resources on the client.

Each microservice instance runs on its own server with its own IP address.

Load Balancing

- Client maintains **cache of service instances** (via Eureka or similar).
- Removes failing instances dynamically.

1. When a service client needs to call a service, it checks a local cache for the service instance IPs. Load balancing between service instances occurs on the service.

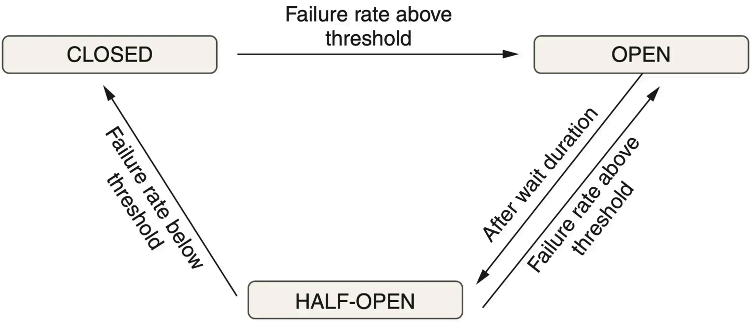

Client applications
Client-side cache/load balancing
Client-side cache/load balancing
3. Periodically, the client-side cache is refreshed with the service discovery layer.
Service discovery layer
Service discovery node 1
Service discovery node 2
Service discovery node 3
Heartbeat
Service instances
X
Service A
2. If the client finds a service IP in the cache, it will use it. Otherwise, it goes to the service discovery.

Benefit: Prevents requests from hitting unhealthy services, reducing cascading failures.

Circuit Breaker

Monitors dependent services and **fails fast** if unhealthy.

- **Open:** block calls, use fallback
- **Half-open:** test if service recovered
- **Closed:** normal operations


CLOSED
Failure rate above threshold
OPEN
Failure rate below threshold
After wait duration
Failure rate above threshold
HALF-OPEN

Fixed Window

- Count failures in fixed intervals (e.g., 10s).
- Open circuit if threshold exceeded.
- **Pros:** Simple. **Cons:** Slow to detect bursts at window edges.

Sliding Window

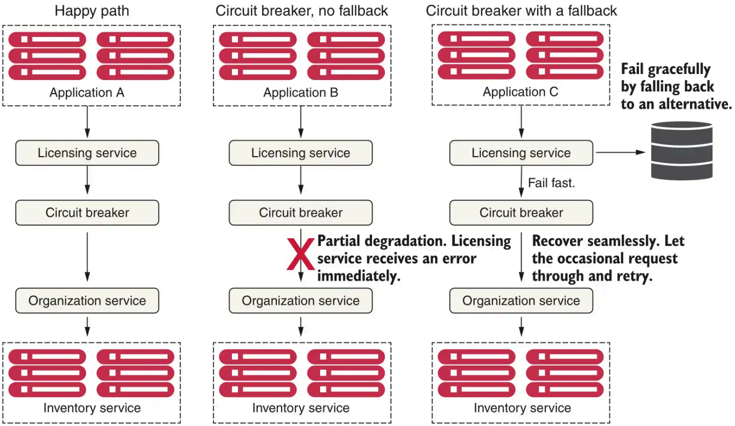
- Continuously track failures over a rolling period.
- Open circuit if failure ratio exceeds threshold.
- **Pros:** Fast detection. **Cons:** Sensitive to short bursts.

Leaky Bucket

- Failures accumulate in a “bucket” that leaks over time.
- Open circuit if bucket fills up.
- **Pros:** Smooths bursts. **Cons:** Needs tuning (bucket size/leak rate).

Fallback

- Execute alternative logic when remote calls fail
- Prevents **exceptions from crashing clients**


Happy path
Circuit breaker, no fallback
Circuit breaker with a fallback
Application A
Application B
Application C
Fail gracefully by falling back to an alternative.
Licensing service
Licensing service
Licensing service
Fail fast.
Circuit breaker
Circuit breaker
Circuit breaker
Partial degradation. Licensing service receives an error immediately.
Recover seamlessly. Let the occasional request through and retry.
Organization service
Organization service
Organization service
Inventory service
Inventory service
Inventory service

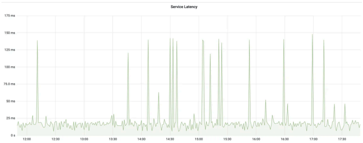
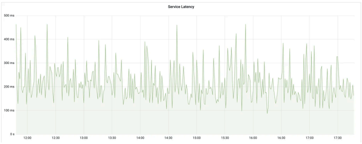
Example: Return cached data or default values if a service is down.

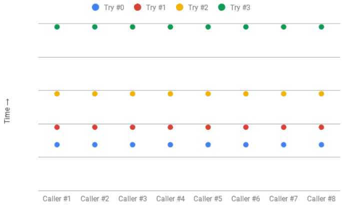
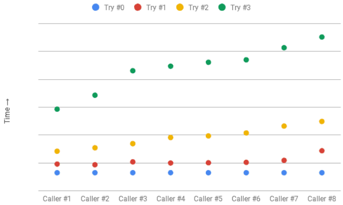
Retry

- Only for **idempotent operations**
- **Exponential backoff + jitter** smooths traffic spikes

⚠️ Network saturation risk: Use **circuit breakers** to prevent **retry storms**

- Retries increase traffic → more congestion
- More congestion → more failures
- More failures → even more retries

Retry suitable	Retry not suitable
	

Backoff	Backoff + jitter
	

Tip: Avoid retries for **time-critical operations**; failing fast (**fallback**) is often better.

Timeout

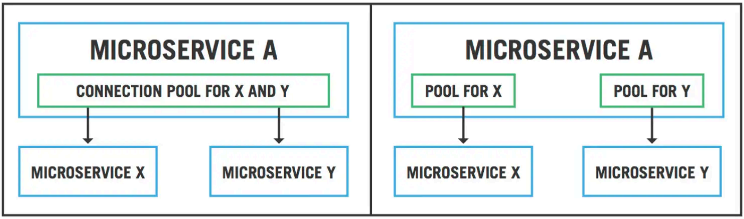
- Set maximum wait for synchronous calls
- **Too high:** wastes resources
- **Too low:** triggers unnecessary retries → latency spikes

Best practice: use **high-percentile latency metrics** (e.g., 99.9th percentile) to choose timeouts.

Bulkhead

Isolate resources per dependency to prevent one slow service from consuming all resources.

- Different thread pools or connection pools per downstream service
- Failures confined → other parts remain operational


MICROSERVICE A
CONNECTION POOL FOR X AND Y
MICROSERVICE X
MICROSERVICE Y
MICROSERVICE A
POOL FOR X
POOL FOR Y
MICROSERVICE X
MICROSERVICE Y

Example: Service A with `/a/x` (slow) and `/a/y` (fast).

Without bulkheads → `/a/x` consumes all threads → `/a/y` blocked.

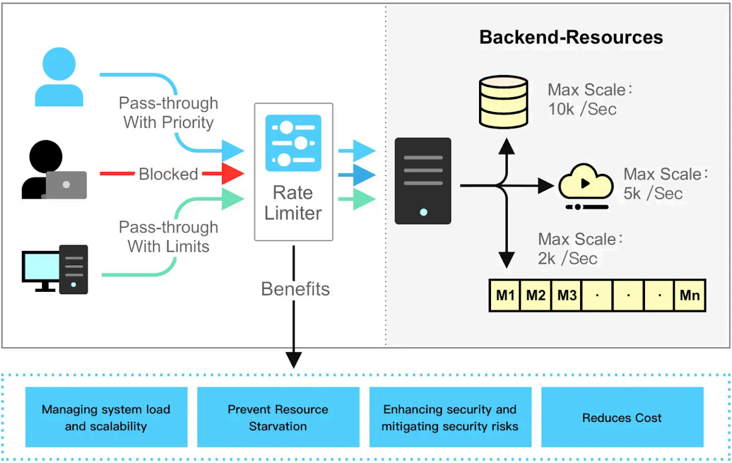
With bulkheads → `/a/x` threads isolated → `/a/y` continues.

Server-Side Resiliency Patterns

Rate Limiter

Controls the **rate of incoming requests**.

- Prevents overload, DoS attacks, and unfair resource usage
- Strategies: fixed window, sliding window



Backend-Resources
Pass-through With Priority
Blocked
Pass-through With Limits
Rate Limiter
Max Scale : 10k /Sec
Max Scale : 5k /Sec
Max Scale : 2k /Sec
M1
M2
M3
·
·
·
Mn
Benefits
Managing system load and scalability
Prevent Resource Starvation
Enhancing security and mitigating security risks
Reduces Cost

Load Shedding vs Rate Limiting:

- **Rate Limiting:** throttle **before processing**
- **Load Shedding:** drop requests **when overloaded**

Resources

- *Microservices with Spring Boot 3 and Spring Cloud* – Chapter 13
- YouTube Video on Microservices Resiliency
- AWS Article: Exponential Backoff and Jitter